Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Senior Frontend Engineer with 8 years of experience architecting high-performance, scalable web applications using React, TypeScript, and Next.js across healthcare, fintech, and AI-driven product environments. Proven track record building reusable design systems, optimizing frontend performance through bundle analysis and code splitting, and delivering conversational UI experiences powered by RAG/LLM systems. Deep expertise in automated testing (Jest, Playwright, Cypress), CI/CD pipelines, build tooling (Webpack, Vite, Rollup), and mentoring engineering teams to deliver with speed and quality.

CORE COMPETENCIES

- **Frontend Architecture:** Scalable component systems, design systems (Radix UI, Storybook), monorepos (Nx, Turborepo), SSR/SSG/ISR, and micro-frontend patterns.
- **Frameworks & Libraries:** React, Next.js, Vue.js, TypeScript, modern JavaScript (ESNext), Redux, Zustand, React Query, Tailwind CSS, and Radix UI.
- **Build Tools & Bundlers:** Webpack, Vite, Rollup, webpack-bundle-analyzer, rollup-plugin-visualizer, and Nx monorepo orchestration.
- **Testing & Quality:** Jest, Enzyme, Playwright, Cypress, Storybook, Allure Reports, ESLint, Prettier, Husky, and CI-driven QA.
- **Performance & Observability:** Core Web Vitals, Lighthouse, React Profiler, Chrome DevTools, code splitting, lazy loading, CDN strategies, Sentry, and Datadog RUM.
- **API Integration:** GraphQL, REST APIs, JWT authentication, role-based access control (RBAC), real-time streaming (SSE, EventSource), and WebSockets.
- **AI & Conversational UI:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Chroma vector search, and real-time conversational interfaces.
- **Cloud & CI/CD:** AWS (CloudFront, S3), Vercel, Docker, GitHub Actions, and automated deployment pipelines.
- **Leadership:** Technical architecture reviews, mentorship, code reviews, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a modular **React Chrome extension**, optimizing workflows for clinical site integrations and improving adoption across product teams.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG compliance), and developer velocity across multiple product teams.
- Co-developed a full-stack **Next.js chat widget** with a **conversational UI** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time through natural language.
- Optimized application performance by implementing **lazy loading, code splitting**, dynamic imports, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools**. Applied memoization and responsive layouts to minimize re-renders across devices.
- Integrated **GraphQL** and **REST API** endpoints for seamless data fetching, and implemented **JWT-based authentication** with **role-based access control** using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), conducting regular code reviews and fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a **35% faster load time** and smoother runtime across devices.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.

- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improved release confidence.
- Utilized **Storybook** for documenting and showcasing **React components**, streamlining change management, enabling visual regression testing, and supporting agile development practices across the team.
- Established engineering standards for code quality and deployment using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, conducting code reviews and fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native browser **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and professional development support.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform with conversational UI. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: NestJS API backend, **Next.js** citizen-facing web app with **React 19** and **22 interactive chart types** (Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built the **conversational UI** frontend with **real-time SSE streaming**, 15s keepalive heartbeats, **Tailwind CSS v4**, dark mode, URL state management with nuqs, and responsive mobile layouts optimized for citizen engagement.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests across 4 projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5), and **Docker Compose** production deployment with multi-stage Dockerfiles and health checks.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*